



feature

Mac mini

Is touted as the world's most energy-efficient desktop computer

thousands of people died from the terrible storms that hit northern Europe as the LIA started — and that was just the beginning. Life likes it warm; consider how much life there is in the tropics, and then look at the poles! So, let's leave this issue aside and now look at the real environmental impacts of modern technology such as PCs and mobile phones.

The throw-away society

For thousands of years, nearly everything we used was made of plant or animal material, or from clay. After these were discarded, the biological materials would always decompose over time (they are bio-degradable), and pottery would crumble back to dust. It mattered not that these things were simply chucked away without a second thought. But we have gradually been developing different materials, and this has accelerated dramatically over the last few decades. However, we have continued to behave as though we could still simply chuck things away without thinking. Many products also now have shorter life-cycles than their equivalents a few decades ago, and so the rate at which many things are discarded has tended to increase. The result is growing mountains of rubbish, and a decreasing amount of it is biodegradable.

What to do with this? Apart from burning there seems to be only one option that humans are following, that of land-fill. If ever there was a case of "sweeping a problem under the carpet", this must be it. The problem is just left for others to solve at some distant point in the future. Land-fill is simply dumping, underground instead of over-ground. This would not be a problem if the waste was biodegradable, but increasingly there are toxic materials in the waste, particularly from computer systems, mobile phones, televisions, batteries and so forth — electronic waste, known as e-waste.

The correct alternative is to recycle the materials in these products, but it is not easy. Recycling collection points in the west mainly have collection cans for paper, clear glass and coloured glass, but not much else. The situation is improving, but the main focus is on these most benign of materials that we throw away. They are worth recycling, but a more important target should be the toxins in the e-waste, and this is the fastest growing portion of normal consumer waste materials world-wide.

Not only would the recycling of e-waste prevent toxic materials from entering the environment, many of the materials are relatively rare and so their re-use would necessitate less mining and other forms of original production.

Unfortunately, much of the dumping of e-waste comes from so-called recycling schemes. A report by the Basel Action Network (BAN) a few years ago highlighted a so-called recycling centre in the Guiyu region of China. Very little actual recycling went on, plumes of acrid smoke wafted into the air, grey-black sludge was dumped alongside the local river. Electronic waste simply piled up, and soil samples registered toxins at rates hundreds of times higher than would cause alerts in the west.

The author of the original report returned a few months ago and stated that "It has gotten far, far worse." This is just one of many sites where the soil, water and air are being contaminated by e-waste, that is supposedly being recycled, and yet is



Wires from computer systems being sorted during the day for later burning at night. Note the family homes right next to the site. ©2006 Basel Action Network (BAN)

effectively simply being dumped. Another case of a problem being swept under the carpet and left for others to sort out. Incidentally, all the photos of e-waste in this article are from the Guiyu site investigated by BAN. See BAN's website for further details of other e-waste sites.

The original report mentioned children playing "amid heaps of ash from burned electronic waste", also people exposed to a carbon black from computer printer cartridges (thought to cause cancer), release of toxic lead and barium from cathode

ray tubes, burning of plastics producing dioxins, heating of printed circuit boards to melt solder, producing inhaled toxic fumes. This lists just the most obvious toxins involved — there are in fact many, many more, and their effect on the environment is disastrous, with people having to carry drinking water in from many miles away, and the air is also poisonous. These toxins can cause brain damage, respiratory and skin problems, kidney disease, cancers, and mutations — and that is just to humans; they also damage plant and animal life.

This is not just happening in a few isolated and distant villages in China. BAN found that even in "New Delhi, children are routinely employed to burn circuit boards — again producing toxic fumes that they cannot help but inhale." And then what happens to the left-over waste, after the solder and perhaps a little gold has been salvaged?

The description in the last few paragraphs refers mainly to electronic waste that people — usually in the west — assume has been sent for proper recycling. What happens to the majority of electronic waste that is not sent for recycling at all? Much of this is simply dumped into landfills, gradually seeping its toxins into the local ground and water table.

Clearly, there are four things that are urgently needed: 1) for proper schemes to be set up world-wide to ensure that every single item of electronic equipment is recycled; 2) for there to be proper monitoring of the recycling to ensure that it is done correctly and is not just a cover for dumping; 3) more efforts need to be made in developing less harmful materials for the manufacture of electronic goods so that they pose less of a threat when no longer needed; and, 4) consumers need to be aware of the dangers of e-waste and take very much greater care when purchasing and disposing of electronic items. Unfortunately, not many companies seem to take this seriously, and when questioned pass the responsibility off by saying that it is the job of government, and so on. But they are not all like this. There is the interesting example set by Nokia.

Nokia and recycling

Nokia started a campaign in India at the beginning of this year to encourage customers in four major cities to recycle their phones and chargers, even if they are from a different brand. Nokia provides collection centres for this purpose, and will roll this out across the whole of India in time. Nokia does not do any recycling itself, but "works with qualified recyclers around the world" — the intention being to ensure that the recycling is done properly and with all the correct benefits. The company also promises that it will work on "managing substances in